

Apply GORICA to EffectLiteR estimates

Axel Mayer, Leonard Vanbrabant, and Rebecca M. Kuiper

24 juli 2026

Contents

Data preparation	1
Example 1 - Adjusted Means	2
Inspect the data	2
Preparation for GORIC(A)	2
Set of hypotheses	3
Apply GORICA	4
Example 2: Average Effects	5
Inspect the data	5
Preparation for GORIC(A)	6
Set of hypotheses	6
Apply GORICA	6

This tutorial shows how one can apply the GORICA – an AIC-type information criterion that can evaluate order-restricted, theory-based hypotheses – to the estimates of the **EffectLiteR** package.

The **EffectLiteR** package uses structural equation modeling to estimate average and conditional effects of a treatment variable on an outcome variable, taking into account multiple continuous and categorical covariates. It automatically generates **lavaan** (Rosseel, 2012) syntax for a multi-group structural equation model, runs the model in **lavaan**, and extracts various average and conditional effects of interest. The main function of the package is **effectLite**.

In this tutorial, it will be shown how you can evaluate order-restricted, theory-based hypotheses regarding average and/or conditional effects, using the **restriktor** and **EffectLiteR** packages in R.

For more information regarding **EffectLiteR**, see:

Mayer, A., Dietzfelbinger, L., Rosseel, Y. & Steyer, R. (2016). The EffectLiteR approach for analyzing average and conditional effects. *Multivariate Behavioral Research*, 51, 374-391.

For (more) information regarding interpreting the GORIC(A) output, see ‘Guidelines_output_GORIC’ (<https://github.com/rebeccakuiper/Tutorials>).

Data preparation

First, load the required libraries (after they have been installed). These libraries contain functions, such as **goric**, that will be used in the R code below. Each time you reopen R, you will have to load the required libraries.

```
## First, install the packages, if you have not done this already:
if (!require("EffectLiteR")) install.packages("EffectLiteR")
if (!require("restriktor")) install.packages("restriktor")
if (!require("psych")) install.packages("psych")
```

```
## Then, load the packages:
library(EffectLiteR)
library(restriktor) # for the goric function
library(psych) # for the function describeBy

# If you want to use restriktor from github: if (!require('pak'))
# install.packages('pak') library(pak) pak::pak('LeonardV/restriktor')
# pak::pkg_install('LeonardV/restriktor') library(restriktor) # for goric
# function
```

Second, it is necessary to load the data.

For this Tutorial, you can use the data available from the **EffectLiteR** package. Several datasets are available by loading the package and here the following two datasets are used:

- example01 (for an adjusted means example)
- nonortho (for an average effects example)

To use the data accordingly, it is recommended to save it in a separate object.

```
# Load the data
data_AdjMeans <- example01
data_AvEffects <- nonortho
```

Example 1 - Adjusted Means

Inspect the data

To inspect the dataset, use the head function:

```
head(data_AdjMeans) # Look at first (6) rows of the data
```

	x	k1	kateg2	z1	z2	z3	dv
9	treat2	male	2	0.57664715	-0.4946821	1.51831377	1.5595743
6	treat2	female	1	-0.05489584	-0.9371025	1.54194766	-0.4078295
7	control	male	2	-1.88284217	-2.1620611	0.01940434	0.1354053
11	treat1	female	2	0.68738815	1.0884689	1.01883391	0.2818120
10	control	female	2	-0.18567233	0.6432084	-0.42386395	-0.3440413
6.1	treat2	female	1	-0.84563945	-1.1052582	-0.22462067	0.5631160

To see a more detailed overview of the data via descriptive statistics split by group variable, use the describeBy function with data_AdjMeans\$x set to be a grouping variable:

```
descrstat <- describeBy(data_AdjMeans$dv, data_AdjMeans$x, mat = TRUE, digits = 3)
descrstat
```

	item	group1	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
X11	1	control	1	686	0.018	0.965	0.034	0.021	0.980	-2.746	2.473	5.219	-0.045	-0.297	0.037
X12	2	treat1	1	615	0.008	1.009	0.012	-0.004	1.015	-2.664	4.183	6.846	0.195	0.264	0.041
X13	3	treat2	1	699	0.050	1.015	0.020	0.041	1.081	-2.960	3.283	6.243	0.110	-0.194	0.038

Preparation for GORIC(A)

Use EffectLiteR

To use the estimates for the adjusted means with GORICA, an object with the main function **effectLite** needs to be created:

```
model_AdjMeans <- effectLite(y = "dv", x = "x", z = c("z1", "z2"), data = data_AdjMeans,
  method = "sem")
```

For input in `effectLite`:

y: Dependent variable (character string). Can be the name of a manifest variable or of a latent variable.

x: Treatment variable (character string) treated as categorical variable.

z: Vector of continuous covariates (character vector). Names of both manifest and latent variables are allowed.

data: A data frame containing the data above.

method: Can be one of `c("sem", "lm")` and indicates which function is used to fit the model.

Optional: Extract information from the EffectLiteR object

The function `goric` can use an `effectlite` object as input. Alternatively, `goric` has another input option for the GORICA as well (that is why the GORICA can easily be applied to a broad range of models): One can enter the (structural) parameter estimates and their covariance matrix. Thus, these must be extracted from the `effectlite` object, here called `model_AdjMeans`.

Note: You can retrieve the parameter names in the `effectLite` object, called 'effectlite_object', by using the following code: `colnames(effectlite_object@results@vcov.def)`.

For this example, we are interested in the adjusted means for the dv and the examined covariates and the covariance between them:

```
# colnames(model_AdjMeans@results@vcov.def) # gives parameter names that can be
# used in the hypotheses
parnames <- c("adjmean0", "adjmean1", "adjmean2") # to have a better overview
est_AdjMeans <- model_AdjMeans@results@est[parnames]
VCOV_AdjMeans <- model_AdjMeans@results@vcov.def[parnames, parnames]

round(est_AdjMeans, 3)
```

```
adjmean0 adjmean1 adjmean2
      0.017      0.006      0.051
```

```
round(VCOV_AdjMeans, 2)
```

```
      adjmean0 adjmean1 adjmean2
adjmean0      0        0        0
adjmean1      0        0        0
adjmean2      0        0        0
```

Set of hypotheses

The hypotheses need to be specified using the parameter names in the `effectLite` object. You can retrieve the parameter names in the `effectLite` object called 'effectlite_object' by using the following code: `colnames(effectlite_object@results@vcov.def)`.

Hypotheses

In the example, the following could be hypothesized regarding the adjusted means (before seeing the data):

```
# colnames(model_AdjMeans@results@vcov.def) # gives parameter names that can be
# used in the hypotheses
H1 <- "adjmean1 < adjmean0 < adjmean2"
```

This hypothesis will be inspected versus its complement, meaning all other possible orderings / hypotheses (thus, excluding the one of interest).

Extra: How to specify restrictions

To evaluate the hypotheses of interest, it is necessary to specify the restrictions in these hypotheses correctly:

- Within the `goric` function, it is possible to use the following operators: `>`, `<`, `=`, `<=`, `>=`, `==` (where the last three denote the same constraint as the first three).
- The `goric` function can deal with:
 - pairwise restrictions separated by a semicolon `;` (e.g., `"beta1 > beta2; beta2 > beta3"`), a comma `,`, or an and-sign `&`.
 - combined restrictions consisting of more than one operator (e.g., `"beta1 > beta2 > beta3"`).

Note that, in the code, one should use the labels of the parameter estimates (so, here: `adjmean0` to `adjmean2`); while you should use population parameters (often denoted by Greek letters) in an article.

Extra: Failsafe / Safeguard hypothesis

To prevent from selecting a weak hypothesis, that is, a hypothesis not supported by the data, one should include a failsafe/safeguard hypothesis. This can be:

- `comparison = "unconstrained"`: This includes all possible hypotheses, thus including the one(s) of interest, which is the default in case of multiple theory-based hypotheses;
- `comparison = "complement"`: This includes all other possible hypotheses, thus excluding the one(s) of interest, which is the default in case of one multiple theory-based hypothesis;
- `comparison = "none"`: In this case, no failsafe is used.

The third option should only be used when you know that all the hypotheses in the set cover all possibilities. Namely, only in that case, (at least) one of the hypotheses contains the truth.

The first should be used when the interest lies in comparing the support for multiple theory-based hypotheses (and is therefore the default when multiple hypotheses are entered). One should then first check whether at least one theory-based hypothesis has more support than the unconstrained. If so, the non-weak hypotheses can be compared to all the other (weak and non-weak) theory-based hypotheses in the set. If not, then all the theory-based hypotheses are weak.

The second, that is, the complement, should be used in case there is one theory-based hypothesis (which is therefore the default when one theory-based hypothesis is entered).

Apply GORICA

GORICA

To apply the GORICA to the EffectLiteR adjusted means estimates, use:

```
# Using the effectlite object:
set.seed(123) # More information below
gorica_AdjMeans <- goric(model_AdjMeans, hypotheses = list(H1 = H1))
```

```
restriktor Message: The covariance matrix of the estimates was obtained via 'vcov()'. This is the biased
```

```
# summary(gorica_AdjMeans)
gorica_AdjMeans
```

```
restriktor (0.7): generalized order-restricted information criterion approximation:
```

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights
1	H1	7.005	1.846	-10.318	0.505	0.692	0.696
2	complement	6.985	2.654	-8.663	0.495	0.308	0.304

Conclusion:

The order-restricted hypothesis 'H1' has 2.29 times more support than its complement.

```
# Using extracted estimates and their covariance matrix set.seed(123) # More
# information below gorica_AdjMeans <- goric(est_AdjMeans, VCOV=VCOV_AdjMeans,
# hypotheses=list(H1=H1)) summary(gorica_AdjMeans) gorica_AdjMeans
```

From the output, one can conclude that the order-restricted hypothesis H_1 has 2.3 times more support than its complement. However, also note that the fit (i.e., loglik) part of H_1 and its complement are close. This indicates that there could be support for a boundary; where one or more inequalities are set to (about-)equalities. One could investigate with the benchmarks function, using 'output_type = "rlw"', whether the loglik weights indeed are close.

For more information, see the guidelines ('Guidelines_output_GORIC.html') and/or the benchmark tutorial on <https://github.com/rebeccakuiper/Tutorials>.

Extra: Seed values

In the calculation of the GORIC, an iterative process is used to calculate the penalty / complexity part. Therefore, one needs to set a seed value using the `set.seed`. This has two advantages:

1. Using the same seed value leads to the same penalty value every time this code is run (also helpful for reproducibility).
2. Using different seed values, allows for sensitivity check on the penalty value. If it is sensitive, then increase number of iterations used in calculation of the penalty.

Example 2: Average Effects

When investigating average effects, almost the same steps need to be performed. For more details, see the description above.

Inspect the data

To inspect the dataset, use the `head` function:

```
head(data_AvEffects) # Look at first (6) rows of the data
```

	y	z	x
1	130.98478	2	2
2	139.44400	2	2
3	91.26276	1	2
4	106.05693	1	2
5	119.74821	1	0
6	91.58844	0	2

Here, it can be seen that there are only 3 variables with one continuous dependent variable (y) and two categorical variables, where x is the independent variable and z the categorical covariate.

To see a more detailed overview of the data via descriptive statistics split by group variable, use the `describeBy` function with `data_AvEffects$x` set to be a grouping variable:

```
descrstat <- describeBy(data_AvEffects$y, data_AvEffects$x, mat = TRUE, digits = 3)
descrstat
```

	item	group1	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis
X11	1	0	1	179	110.638	25.604	112.538	111.346	21.500	32.659	176.557	143.898	-0.267	0.586
X12	2	1	1	176	99.418	20.498	98.627	99.148	20.406	40.746	146.339	105.592	0.071	-0.376
X13	3	2	1	145	119.208	29.991	120.077	119.423	33.784	49.264	183.064	133.799	-0.098	-0.554

Preparation for GORIC(A)

Use EffectLiteR

To use the estimates for the average effects with GORICA, an object with the main function `effectLite` needs to be created:

```
model_AvEffects <- effectLite(y = "y", x = "x", k = "z", data = data_AvEffects, method = "sem")
```

Note that now the argument `z` is no longer used, but the argument `k` is:

`k`: Vector of manifest variables treated as categorical covariates (character vector).

Optional: Extract information from the EffectLiteR object

For this example, we are interested in the average effect between the two categorical variables:

```
# colnames(model_AvEffects@results@vcov.def) # gives parameter names that can
# be used in the hypotheses
parnames <- c("Eg1", "Eg2")
est_AvEffects <- model_AvEffects@results@est[parnames]
VCOV_AvEffects <- model_AvEffects@results@vcov.def[parnames, parnames]

est_AvEffects
```

```
      Eg1      Eg2
0.8802244 6.0841473
```

```
VCOV_AvEffects
```

```
      Eg1      Eg2
Eg1 6.784013 4.846823
Eg2 4.846823 10.614931
```

Set of hypotheses

In the example, the following could be hypothesized regarding the average effects (before seeing the data):

```
# colnames(model_AvEffects@results@vcov.def) # gives parameter names that can
# be used in the hypotheses
H1 <- "Eg1 < Eg2"
# Here, we say that the first average effect is smaller than the second average
# effect
```

This hypothesis will be inspected versus its complement, meaning all other possible orderings / hypotheses (thus, excluding the one of interest).

Apply GORICA

To apply the GORICA to the EffectLiteR average effect estimates, use:

```
# Using the effectlite object:
set.seed(123) # More information below
gorica_AvEffects <- goric(model_AvEffects, hypotheses = list(H1 = H1))
```

restriktor Message: The covariance matrix of the estimates was obtained via 'vcov()'. This is the biased

```
# summary(gorica_AvEffects)
gorica_AvEffects
```

restriktor (0.7): generalized order-restricted information criterion approximation:

Results:

	model	loglik	penalty	gorica	loglik.weights	penalty.weights	gorica.weights
1	H1	-3.779	1.500	10.558	0.853	0.500	0.853
2	complement	-5.536	1.500	14.072	0.147	0.500	0.147

Conclusion:

The order-restricted hypothesis 'H1' has 5.80 times more support than its complement.

```
# Using extracted estimates and their covariance matrix set.seed(123)
# gorica_AvEffects <- goric(est_AvEffects, VCOV=VCOV_AvEffects,
# hypotheses=list(H1=H1)) summary(gorica_AvEffects) gorica_AvEffects
```

From the output, one can conclude that the order-restricted hypothesis H_1 has 5.8 times more support than its complement. Notably, since the fit parts clearly differ, there is no indication for support for the boundary (see the 'Guidelines_output_GORIC.html' on <https://github.com/rebeccakuiper/Tutorials>).